

Introduction to Contract Theory and Auctions

Screening with a continuum of types

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Standard Model with a Continuum of Types

- Type set: $\Theta = [\theta_L, \theta_H]$
- Cumulative distribution function of $\theta : F$, continuous density $f(\theta) > 0$.
- $v_1, v_2, v_{12} > 0, v_{11} \leq 0$
- A contract is a pair of functions $(q, t) : \theta \rightarrow \mathbb{R}_+ \times \mathbb{R}$

Continuum of types

The monopolist problem is

$$\begin{aligned} \max_{(q,t)} \int_{\theta_L}^{\theta_H} (t(\theta) - c(q(\theta)))f(\theta)d\theta \\ \text{subject to} \\ v(q(\theta), \theta) - t(\theta) \geq v(q(\theta'), \theta) - t(\theta'), \forall \theta, \theta' \\ v(q(\theta), \theta) - t(\theta) \geq 0, \forall \theta \end{aligned}$$

For any contract (q, t) let

$$U(\theta) = v(q(\theta), \theta) - t(\theta)$$

We first characterize incentive compatible contracts

Theorem 1

A contract (q, t) is incentive compatible if and only if

- 1** *q is increasing*
- 2** *$U(\theta) = U(\theta') + \int_{\theta'}^{\theta} v_2(q(\tau), \tau)d\tau$ for any θ, θ' .*

Proof

⇒ We can write IC constraints of types $\theta' > \theta$ as

$$\begin{aligned}U(\theta) &\geq U(\theta') + v(q(\theta'), \theta) - v(q(\theta'), \theta') \\U(\theta') &\geq U(\theta) + v(q(\theta), \theta') - v(q(\theta), \theta)\end{aligned}$$

Therefore

$$v(q(\theta), \theta') - v(q(\theta), \theta) \leq U(\theta') - U(\theta) \leq v(q(\theta'), \theta') - v(q(\theta'), \theta)$$

- Single crossing then implies $q(\theta') \geq q(\theta)$ (verify)
- Monotonicity of q implies that it is continuous almost everywhere.

Proof

Also

$$\frac{v(q(\theta), \theta') - v(q(\theta), \theta)}{\theta' - \theta} \leq \frac{U(\theta') - U(\theta)}{\theta' - \theta} \leq \frac{v(q(\theta'), \theta') - v(q(\theta'), \theta)}{\theta' - \theta}$$

Taking limits as $\theta' \rightarrow \theta$

$$\frac{dU(\theta)}{d\theta} = v_2(q(\theta), \theta)$$

at all points where q is continuous. If U is absolutely continuous (a strong form of continuity), then

$$U(\theta) = U(\theta') + \int_{\theta'}^{\theta} v_2(q(\tau), \tau) d\tau \quad \forall \theta, \theta' \quad (1)$$

This **integral** form of characterizing incentive compatibility is actually a result from a more general type of results called **envelope theorems**. The particular version that applies here is quite powerful. Please see Milgrom and Segal (2002) Econometrica paper for the original result, Milgrom's textbook (in the syllabus), chapter 3, and my own brief notes on the topic (will upload to Moodle later this week).

Proof

⇐ Consider any $\theta' > \theta$.

$$\begin{aligned}U(\theta') - U(\theta) &= \int_{\theta}^{\theta'} v_2(q(\tau), \tau) d\tau \\&\geq \int_{\theta}^{\theta'} v_2(q(\theta), \tau) d\tau \\&= v(q(\theta), \theta') - v(q(\theta), \theta)\end{aligned}$$

where the inequality follows from SCP and q increasing.

Solving the Model

Also note that

$$\begin{aligned} u(\theta) &= u(\theta_L) + \int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau \\ &\geq u(\theta_L) + \int_{\theta_L}^{\theta} v_2(q(\theta_L), \tau) d\tau \\ &= u(\theta_L) + v(q(\theta_L), \theta) - v(q(\theta_L), \theta_L) \\ &\geq 0 \end{aligned}$$

again by SCP, the fact that q is increasing, and $v_2 > 0$. Therefore, we need only to consider PC constraint for type θ_L .

Solving the Model

Therefore, the problem is equivalent to

$$\max_{(q,t)} \int_{\theta_L}^{\theta_H} (t(\theta) - c(q(\theta))) f(\theta) d\theta \quad (2)$$

$$\text{subject to} \quad (3)$$

$$U(\theta) = U(\theta_L) + \int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau \quad \forall \theta \quad (4)$$

$$q \text{ is increasing} \quad (5)$$

$$U(\theta_L) \geq 0 \quad (6)$$

Using (4) we can write the objective function as

$$\int_{\theta_L}^{\theta_H} \left[v(q(\theta), \theta) - c(q(\theta)) - \int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau \right] f(\theta) d\theta - U(\theta_L)$$

Integration by parts gives [Let $u = \int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau$]

$$\int_{\theta_L}^{\theta_H} \left[\int_{\theta_L}^{\theta} v_2(q(\tau), \tau) d\tau \right] f(\theta) d\theta = \int_{\theta_L}^{\theta_H} (1 - F(\theta)) v_2(q(\theta), \theta) d\theta$$

Therefore, the objective function can be written as

$$\int_{\theta_L}^{\theta_H} \left[v(q(\theta), \theta) - c(q(\theta)) - \frac{1 - F(\theta)}{f(\theta)} v_2(q(\theta), \theta) \right] f(\theta) d\theta - U(\theta_L)$$

Therefore, the monopolist's problem reduces to

$$\max_q \int_{\theta_L}^{\theta_H} \left[v(q(\theta), \theta) - c(q(\theta)) - \frac{1 - F(\theta)}{f(\theta)} v_2(q(\theta), \theta) \right] f(\theta) d(\theta)$$

subject to q increasing.

Under some regularity assumptions on the objective function (ask me if you are curious), we can differentiate under the integral sign to get the first order condition with respect to q for each θ :

$$v_1(q(\theta), \theta) - c'(q(\theta)) - \frac{1 - F(\theta)}{f(\theta)} v_{12}(q(\theta), \theta) = 0$$

Note that the monopolist maximizes virtual surplus!

- No distortion at the top:

$$v_1(q(\theta), \theta) = c'(q(\theta))$$

- Lower types under-consume since

$$\frac{1 - F(\theta)}{f(\theta)} v_{12}(q(\theta), \theta) > 0$$

- All types other than the lowest receive information rent.
- If virtual surplus function

$$v(q, \theta) - c(q) - \frac{1 - F(\theta)}{f(\theta)} v_2(q, \theta)$$

is strictly concave in q , then the solution is unique.

Monotonicity

If

- $\frac{1-F(\theta)}{f(\theta)}$ is decreasing (monotone hazard rate property)
- $\frac{\partial v_{12}}{\partial \theta} \leq 0$

then q is increasing. To see this, let θ' but $q(\theta') < q(\theta)$. Then

$$\begin{aligned} 0 &= v_1(q(\theta), \theta) - c'(q(\theta)) - \frac{1-F(\theta)}{f(\theta)} v_{12}(q(\theta), \theta) \\ &\leq v_1(q(\theta'), \theta) - c'(q(\theta')) - \frac{1-F(\theta)}{f(\theta)} v_{12}(q(\theta'), \theta) \\ &\leq v_1(q(\theta'), \theta') - c'(q(\theta')) - \frac{1-F(\theta)}{f(\theta)} v_{12}(q(\theta'), \theta) \\ &\leq v_1(q(\theta'), \theta') - c'(q(\theta')) - \frac{1-F(\theta')}{f(\theta')} v_{12}(q(\theta'), \theta') \\ &= 0 \end{aligned}$$

which is a contradiction.

Summary

Common results from the standard screening model:

- At the efficient allocation, high types want to mimic low types → first best is not incentive compatible.
- They have to receive (information) rents to prevent mimicking.
- At second best principal trades off efficiency for information rents.
 - ▶ Less than efficient for low types to reduce rent for high types.
 - ▶ Maximizes virtual surplus = surplus - information rent.
- Efficient consumption for the highest type.
- All types except the lowest receive information rent.
- Higher types consume more than lower ones.

Extensions?

- Multidimensional q and θ : we have seen a model with multidimensional θ . However, the case for multidimensional q and θ , except in special cases is extremely difficult to solve.
- Principal's payoff directly depends on θ : common values.
- Agent's reservation utility depends on θ .
- Multi-principal and/or multi-agent models (If we have time, I will talk about auctions).
- Dynamic models.